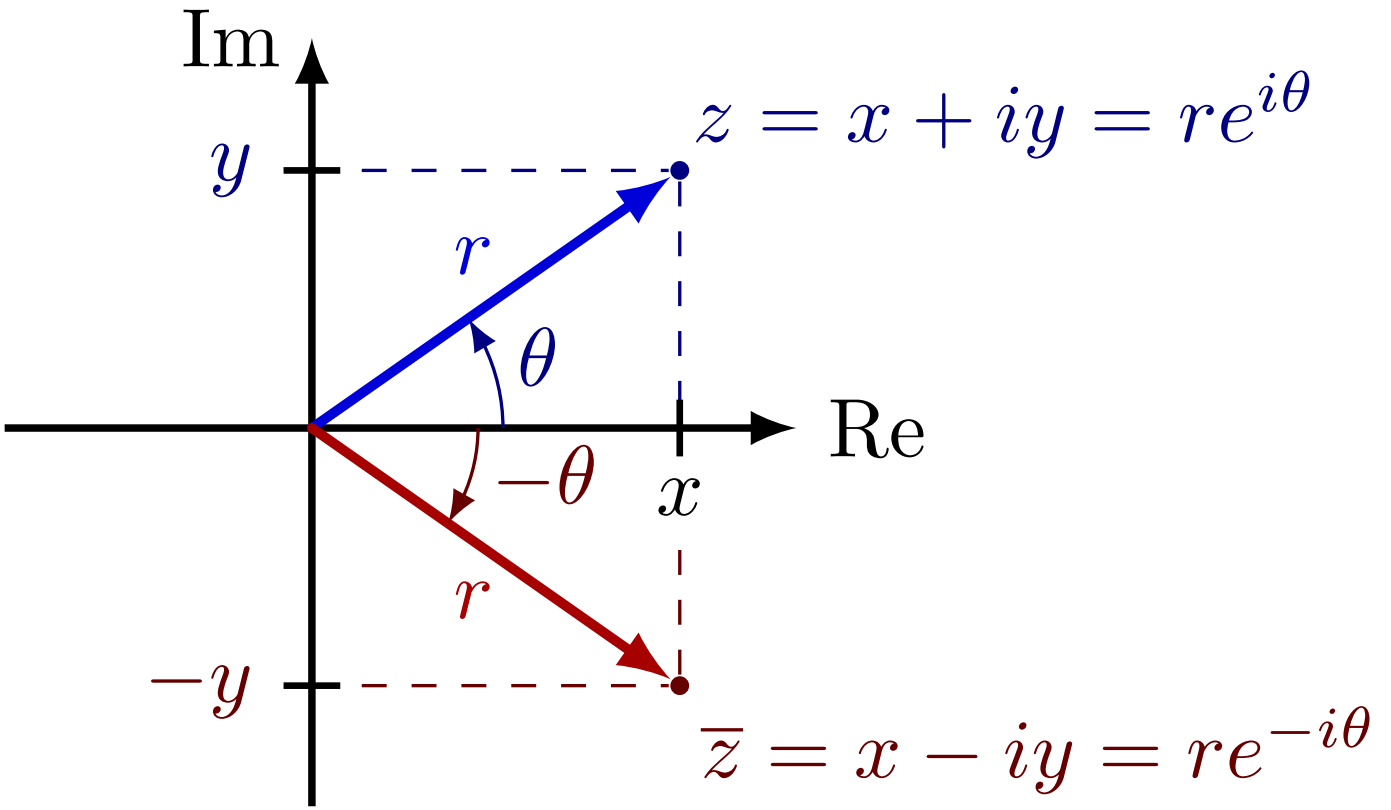
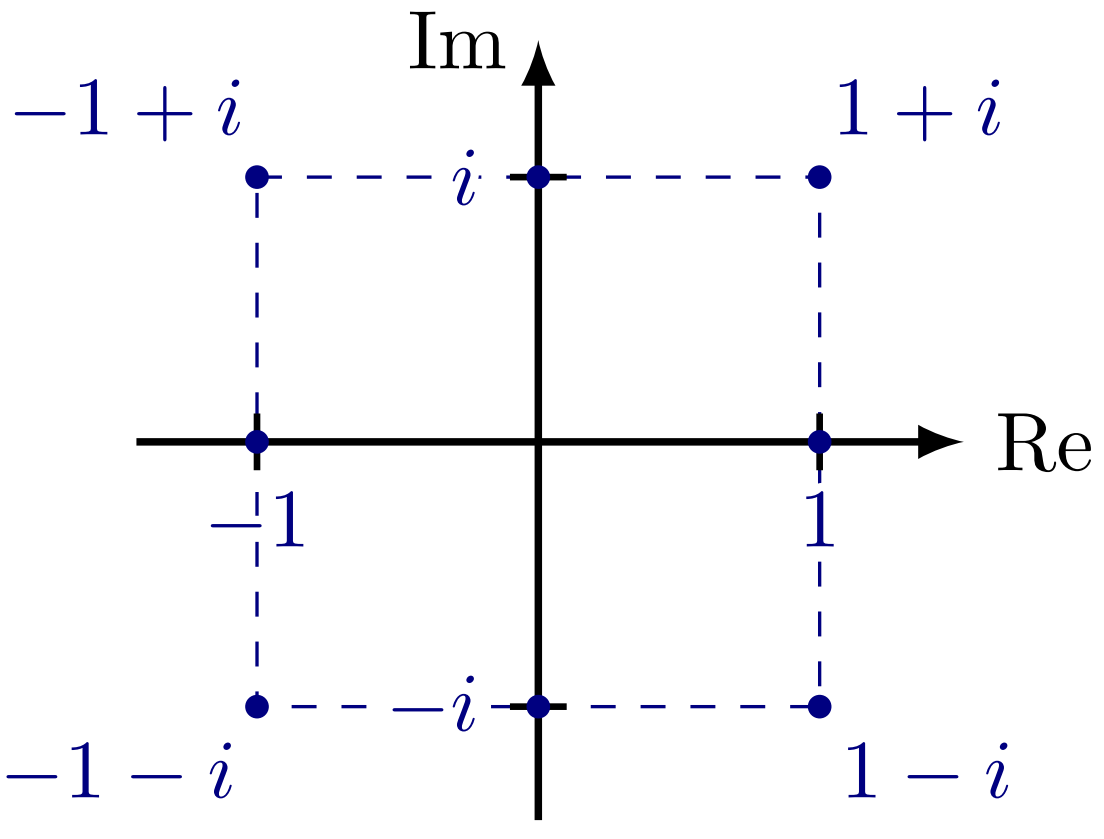
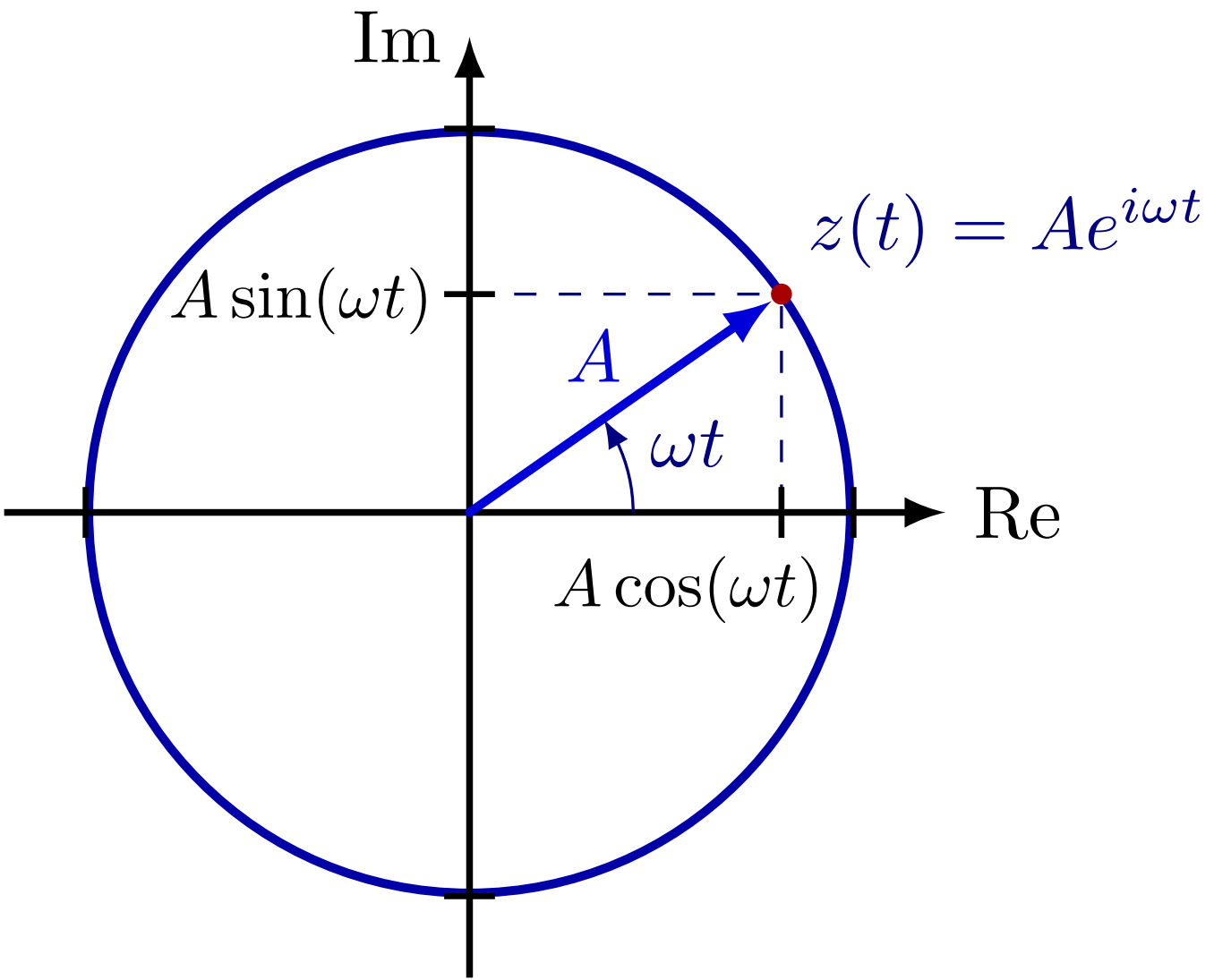
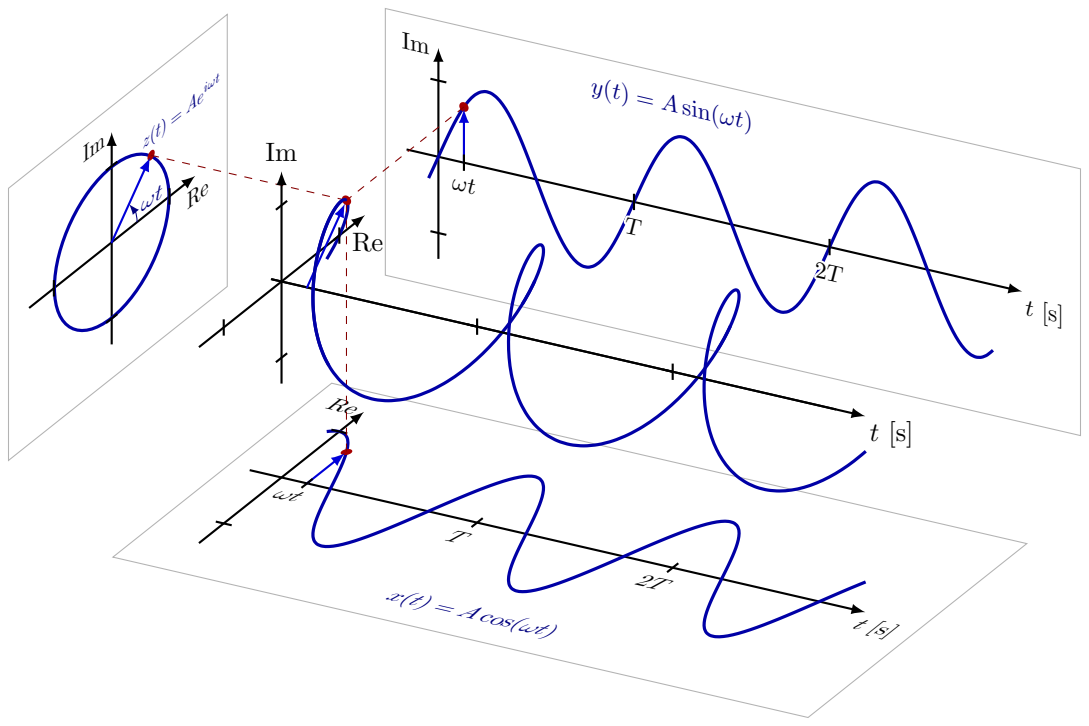
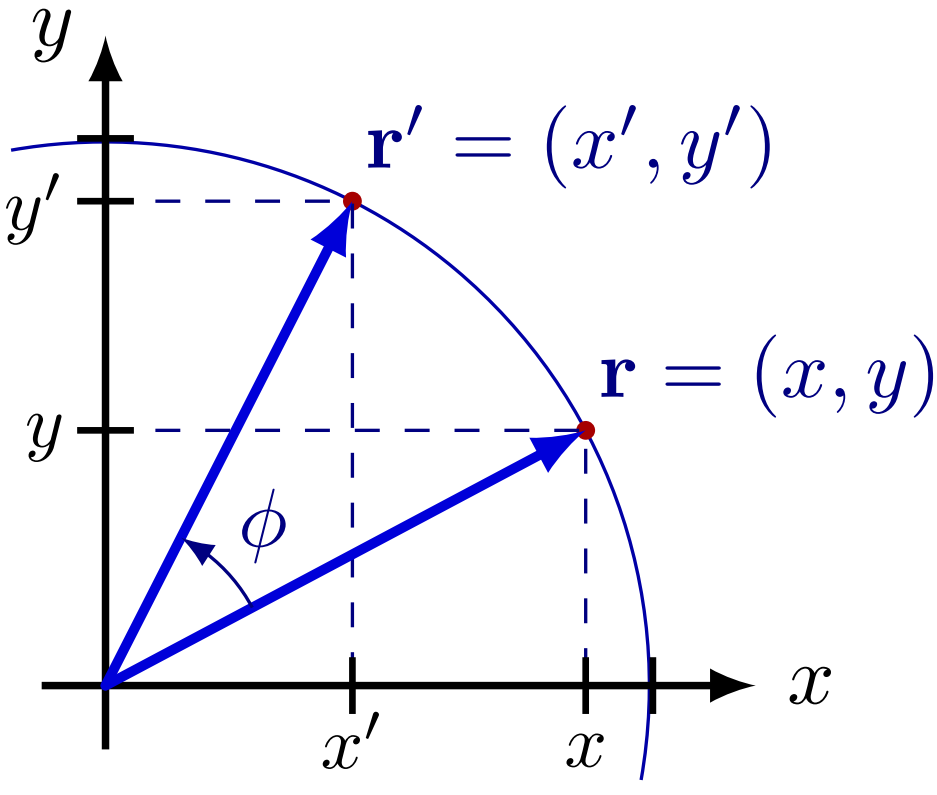
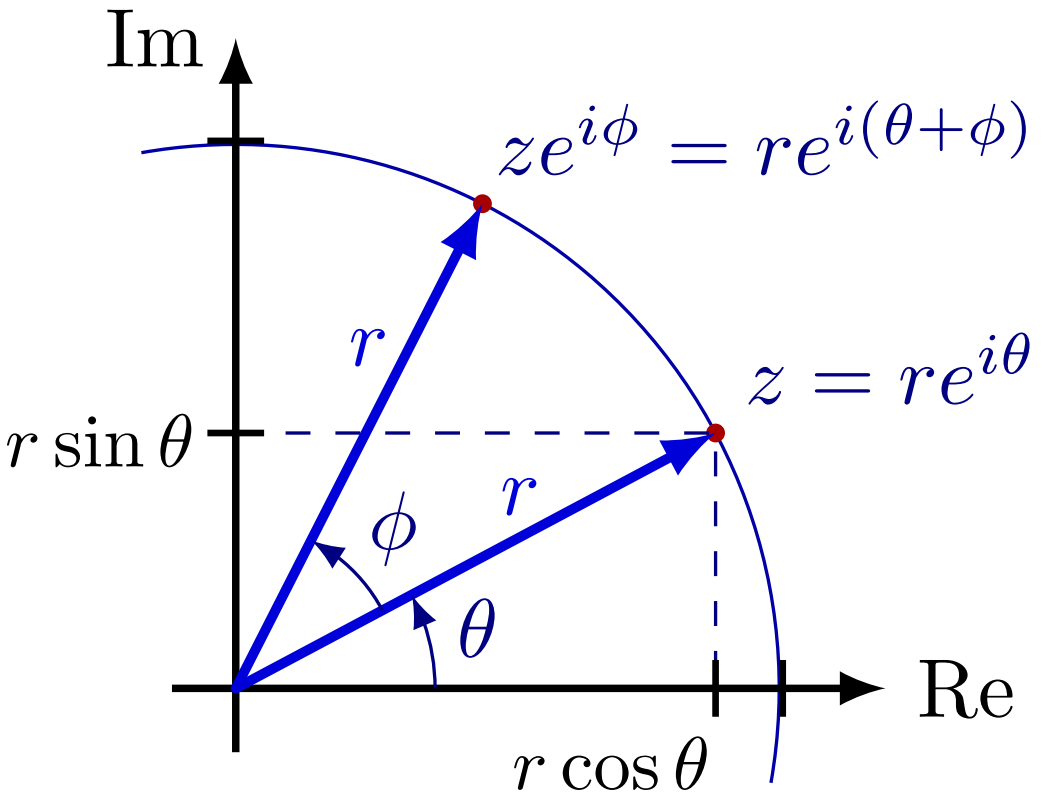

Im
Re
y
x
$-y$
r
θ
$-\theta$
r
$z = x + iy = re^{i\theta}$
$\overline{z} = x - iy = re^{-i\theta}$


Im
Re
$-1+i$
i
$1+i$
-1
1
$-i$
$-1-i$
$1-i$


Im
Re
$z(t) = Ae^{i\omega t}$
$A\sin(\omega t)$
A
ωt
$A\cos(\omega t)$


Im
Re
$z(t) = Ae^{i\omega t}$
ωt
Im
Re
Im
$y(t) = A\sin(\omega t)$
ωt
T
$2T$
t [s]
t [s]
Re
ωt
T
$2T$
$x(t) = A\cos(\omega t)$
t [s]


y
$\mathbf{r}' = (x', y')$
y'
$\mathbf{r} = (x, y)$
y
ϕ
x
x'
x


Im
$ze^{i\phi} = re^{i(\theta+\phi)}$
$z = re^{i\theta}$
r
$r \sin\theta$
r
ϕ
θ
Re
$r \cos\theta$